UNIT 3 – LINKED LIST

TOPIC - SMART CANTEEN ORDER MANAGEMENT SYSTEM

1. Introduction

A college canteen receives many food orders from students every day. Managing these orders properly helps the canteen staff keep track of pending and completed orders.

The Smart Canteen Order Management System is a menu-driven C++ program that uses a Singly Linked List to store and manage student food orders dynamically.

The system allows users to insert orders at the beginning or end, display all pending orders, search for a specific order, and delete completed orders.

2. Problem Statement

Implement a Singly Linked List to maintain student food orders. Perform insertion at the beginning and end, deletion of completed orders, searching for a specific order, and displaying all pending orders.

3. Objectives

1. To store student food orders dynamically.
2. To insert a new order at the beginning or end of the list.
3. To display all pending canteen orders.
4. To search for a particular order using its Order ID.
5. To delete an order after it is completed.
6. To understand the practical implementation of a Singly Linked List using C++.

4. Data Structure Used

Singly Linked List

A Singly Linked List is a linear data structure consisting of nodes. Each node contains data and a pointer to the next node.

In this project, each node stores:

- Order ID
- Student Name
- Food Item
- Pointer to the next node

The `head` pointer stores the address of the first node. The last node points to `NULL`.

Node Structure

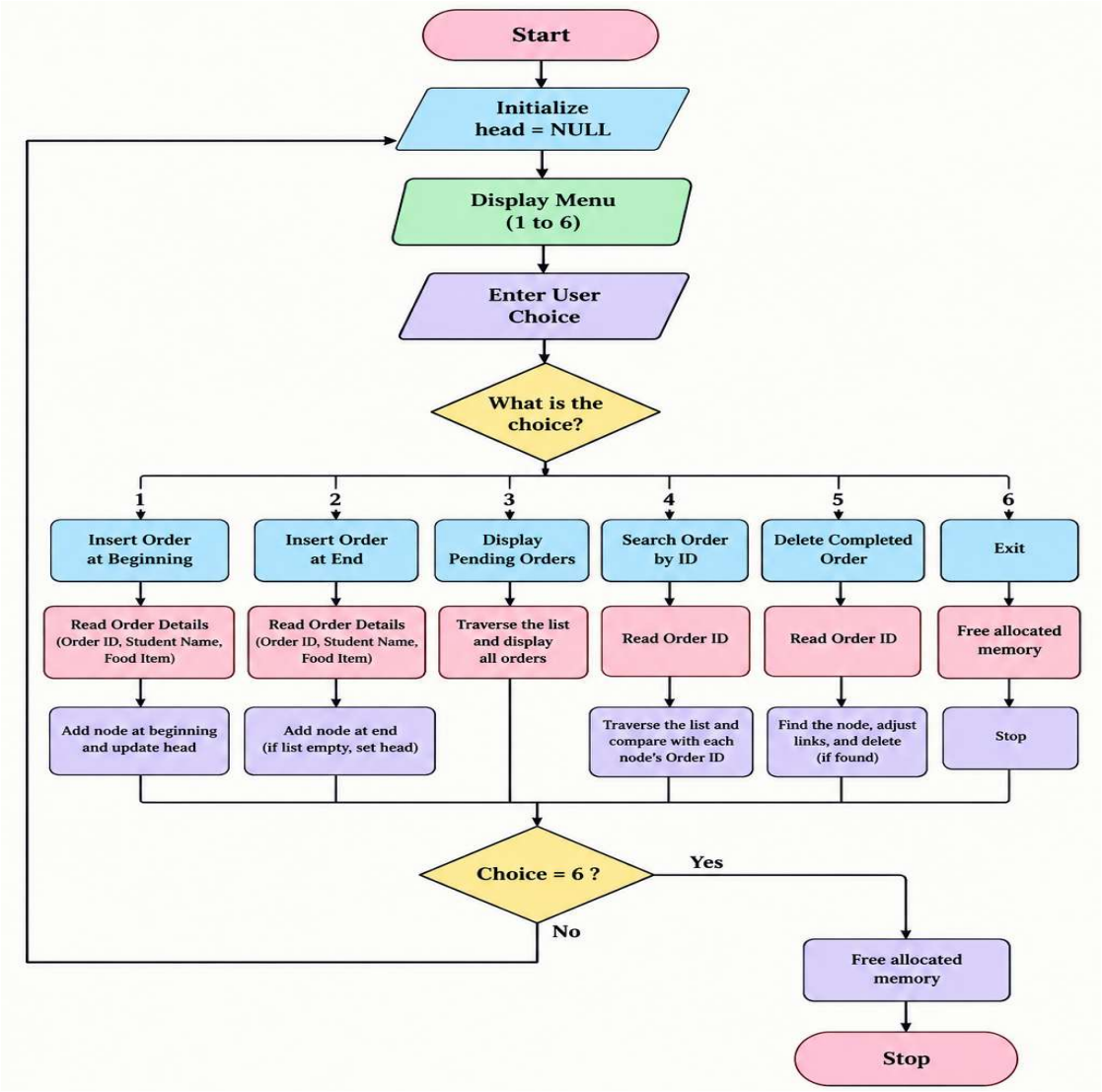
```
[Order Details | Next Pointer] → [Order Details | Next Pointer] → NULL
```

The list can grow or shrink dynamically as orders are added or deleted.

5. Algorithm

1. Start the program.
2. Initialize the head pointer to NULL.
3. Display the menu of operations.
4. Accept the user's choice.
5. If the choice is 1, insert an order at the beginning.
6. If the choice is 2, insert an order at the end.
7. If the choice is 3, display all pending orders.
8. If the choice is 4, search for an order using its Order ID.
9. If the choice is 5, delete a completed order using its Order ID.
10. If the choice is 6, exit the program.
11. Repeat the menu until the user selects Exit.
12. Release the allocated memory and stop the program.

6. Flowchart



7. Working Of The Project

The program displays a menu that allows the user to perform different operations on canteen orders.

7.1 Insertion at Beginning

A new order is created and linked before the current first node. The head pointer is updated to point to the new node.

7.2 Insertion at End

A new order is created and added after the last node. If the list is empty, the new node becomes the first node.

7.3 Display Pending Orders

The program traverses the linked list from the head node to the last node and displays the Order ID, student name, and food item for each order.

7.4 Searching an Order

The program asks the user to enter an Order ID. It traverses the list and compares the entered ID with each node's Order ID. If a match is found, the order details are displayed.

7.5 Deleting a Completed Order

The program searches for the specified Order ID and removes the corresponding node by adjusting the links. The memory occupied by the deleted node is released.

8. Programming Language And Concepts

Programming Language: C++

Concepts Used:

- Structures
- Pointers
- Singly Linked List
- Dynamic Memory Allocation
- Insertion and Deletion
- Searching and Traversal
- Functions
- Switch Statement
- Loops and Conditional Statements

9. Advantages

1. **Dynamic Size:** The list can grow or shrink according to the number of orders.
2. **Easy Insertion:** New orders can be added at the beginning or end.
3. **Easy Deletion:** Completed orders can be removed without shifting other elements.

4. **Searching:** Orders can be searched using their Order ID.
5. **Memory Management:** Memory is allocated when a new order is created and released when an order is deleted.
6. **Simple Interface:** The menu makes the program easy to operate.

10. Applications

1. College canteen order management.
2. Small cafeteria order tracking.
3. Food counter order handling.
4. Basic restaurant order record systems.
5. Learning linked-list operations in data structures.

11. Conclusion

We successfully implemented the Smart Canteen Order Management System using a Singly Linked List in C++. The program performs insertion at the beginning and end, displays pending orders, searches for a specific order, and deletes completed orders. This project demonstrates the practical use of linked lists for dynamic data management.